ICLR

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Efficient and accurate explanation estimation with distribution compression

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TL;DR: Efficient data subsampling improves the explanation approximation error of feature attributions, importance, and effects.

Open Review .net

1. Motivation: Sampling from the dataset is prevalent in various post-hoc explanation algorithms, e.g. SHAP, SAGE, PDP, and Expected Gradients. Sampling decreases the **computational cost** of estimation but introduces an **approximation error** and **variance**.

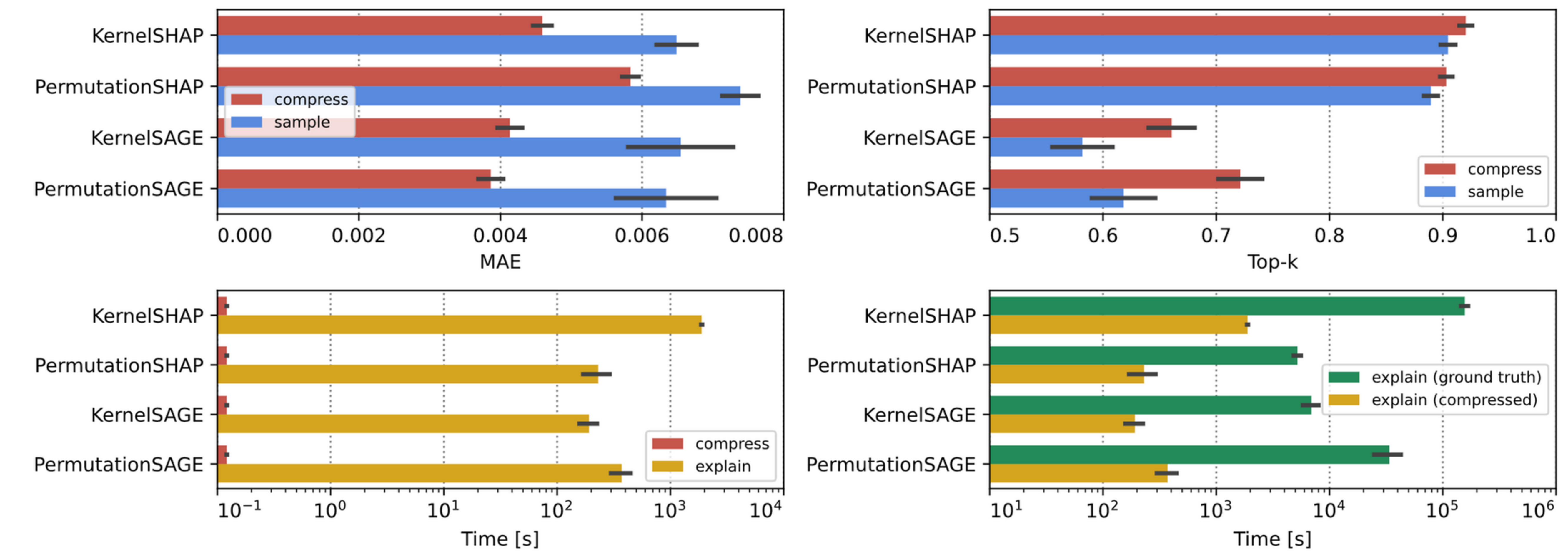
2. Problem: How large is the error? Is it significant? **Can we reduce it?**

3. Yes, we can! Error is sometimes large; it can impact the ranking of feature importance.

4. Solution: Distribution compression, e.g. the kernel thinning algorithm.

(Shetty, Dwivedi & Mackey, ICLR 2022)

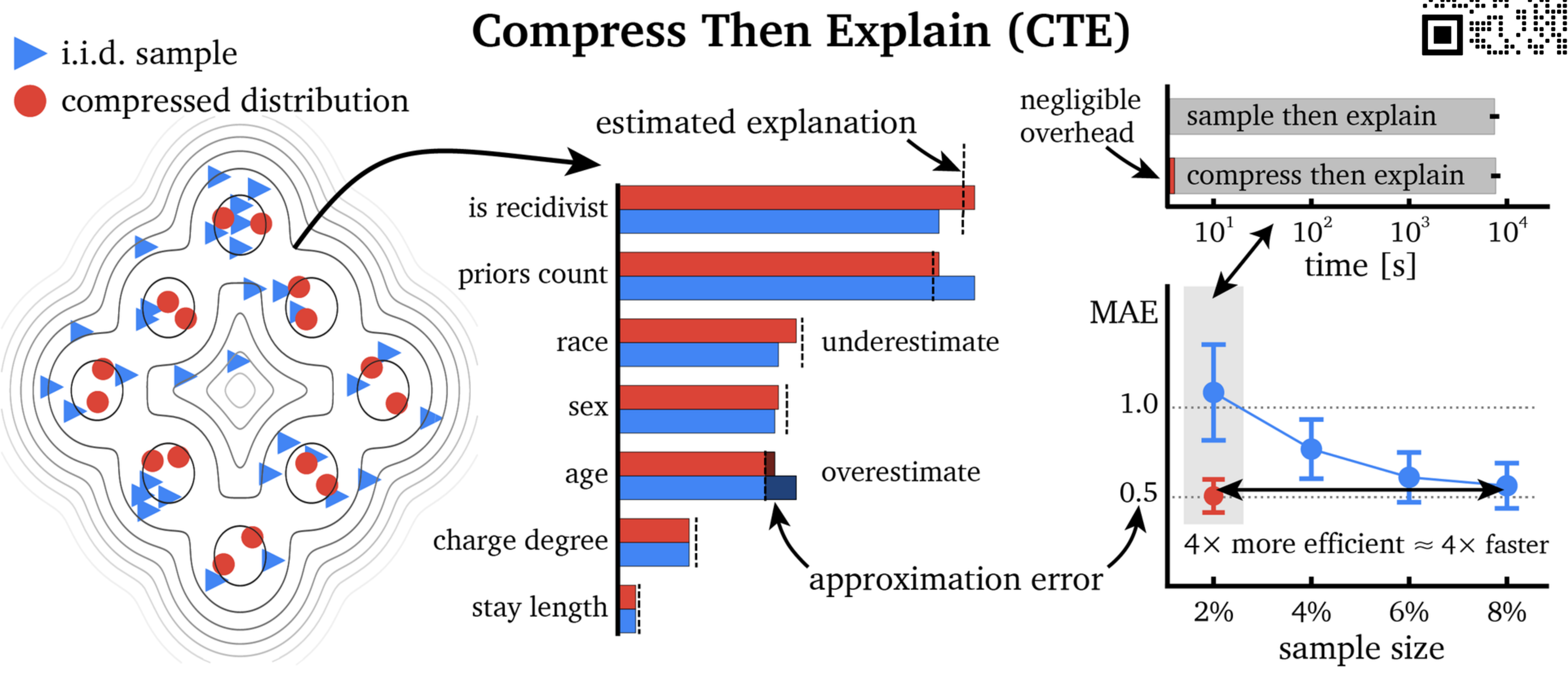
5. Results:

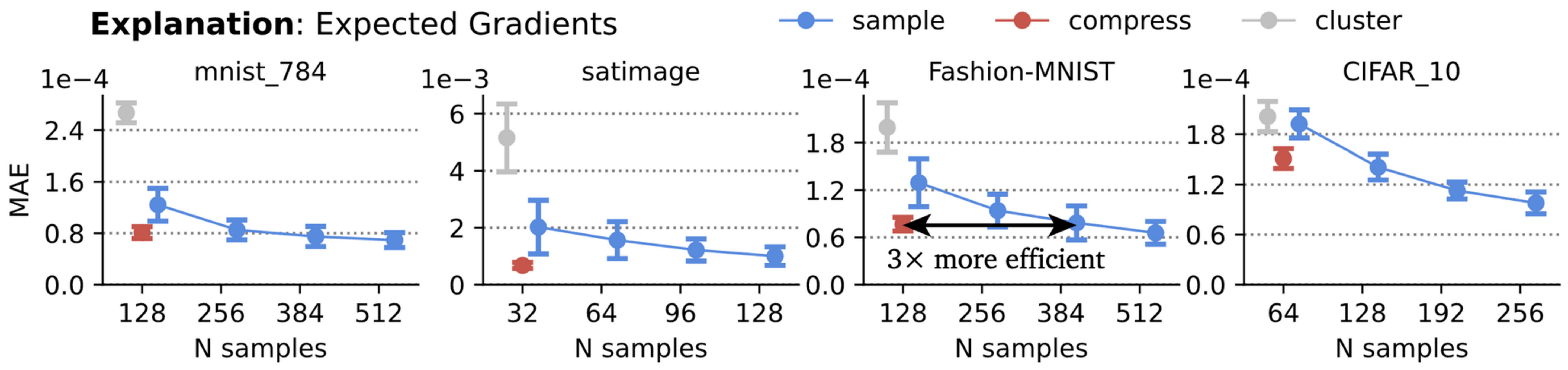


6. Why? Intuition:

Proposition 1 (Feature marginalization is bounded by the maximum mean discrepancy between data samples). *For two empirical distributions $q_{\mathbb{X}}, q_{\tilde{\mathbb{X}}}$ approximated with a kernel density estimator* $\mathbf{k}$, *we have* $\left|f(\mathbf{x}_s; q_{\mathbb{X}}) - f(\mathbf{x}_s; q_{\tilde{\mathbb{X}}})\right| \leq C_f \cdot \widehat{\mathrm{MMD}}_{\mathbf{k}}(q_{\mathbb{X}}, q_{\tilde{\mathbb{X}}})$, *where* C_f *denotes a constant that bounds the model function* f, *i.e.* $\forall_{\mathbf{x}\in\mathbb{R}^p} \left|f(\mathbf{x})\right| \leq C_f$.

Compress Then Explain (CTE)





Compress Then Explain (CTE): not only 2-3x more efficient, but also more stable!